

ECO102: Introduction to Macroeconomics

Lecture 7: Economic Growth Contd.

Wasama Ahmed Khan

BRAC University

Fall 2025

Outline

- ❑ Growth of Potential GDP: Labour Productivity
- ❑ Preconditions for Labour Productivity Growth
- ❑ Policies for Achieving Faster Growth

Growth of Potential GDP: Labour Productivity

- ***Labour productivity*** is the quantity of real GDP produced by an hour of labor
- It is calculated by dividing ***real GDP*** by ***aggregate labour hours***
- When labor productivity grows, real GDP per person grows and brings a rising standard of living

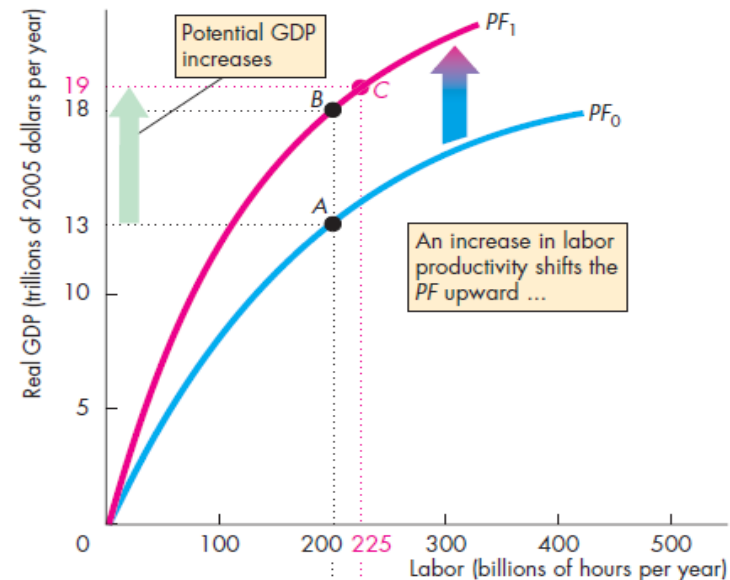
HOW DOES THE POTENTIAL GDP CHANGE DUE TO LABOUR SUPPLY AND HOW DOES IT CHANGE DUE TO LABOUR PRODUCTIVITY?

Growth of Potential GDP: Labour Productivity

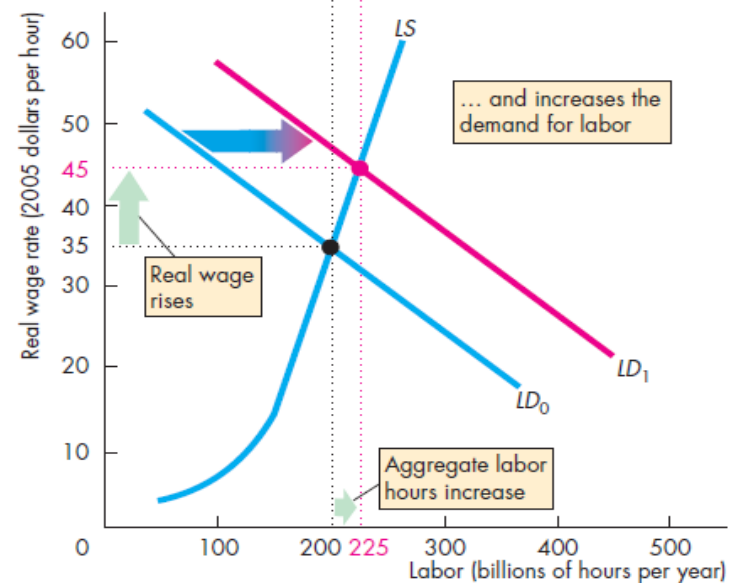
- ***The Effects of an Increase in Labour Productivity:*** An increase in Labour Productivity, expands the production possibilities frontier
- The quantity of real GDP that any given quantity of labour can produce increases as well
- If labour is more productive, firms are willing to pay more for a given number of hours of labour so the demand for labour also increases
- With an ***increase*** in the demand for labour and ***no change*** in the supply of labour, the real wage rate ***rises*** and consequently the quantity of labour supplied ***increases***, increasing the equilibrium quantity of labour
- So an increase in labor productivity increases potential GDP for two reasons: Labour is more productive and more labour is employed

An increase in labour productivity shifts the production function upward from PF_0 to PF_1 in part (a), shifting the demand for labour curve rightward from LD_0 to LD_1 in part (b)

The increase in labour demand, increases the real wage rate, increasing the equilibrium labour hours. The increase in labour hours further increases the potential GDP



(a) Potential GDP



(b) The labor market

Preconditions for Labour Productivity Growth

- Labour productivity is the key to increasing output per hour of labor and rising living standards; the source of the increase in potential GDP is an increase in labour productivity
- The fundamental precondition for labour productivity growth is the incentive system created by firms, markets, property rights, and money
- With the preconditions for labor productivity growth in place, three things influence its pace:
 - Physical Capital Growth
 - Human Capital Growth
 - Technological Advances

Preconditions for Labour Productivity Growth

- ***Physical Capital Growth:*** With access to large amount of capital, labour is able to produce more goods and services, increasing output

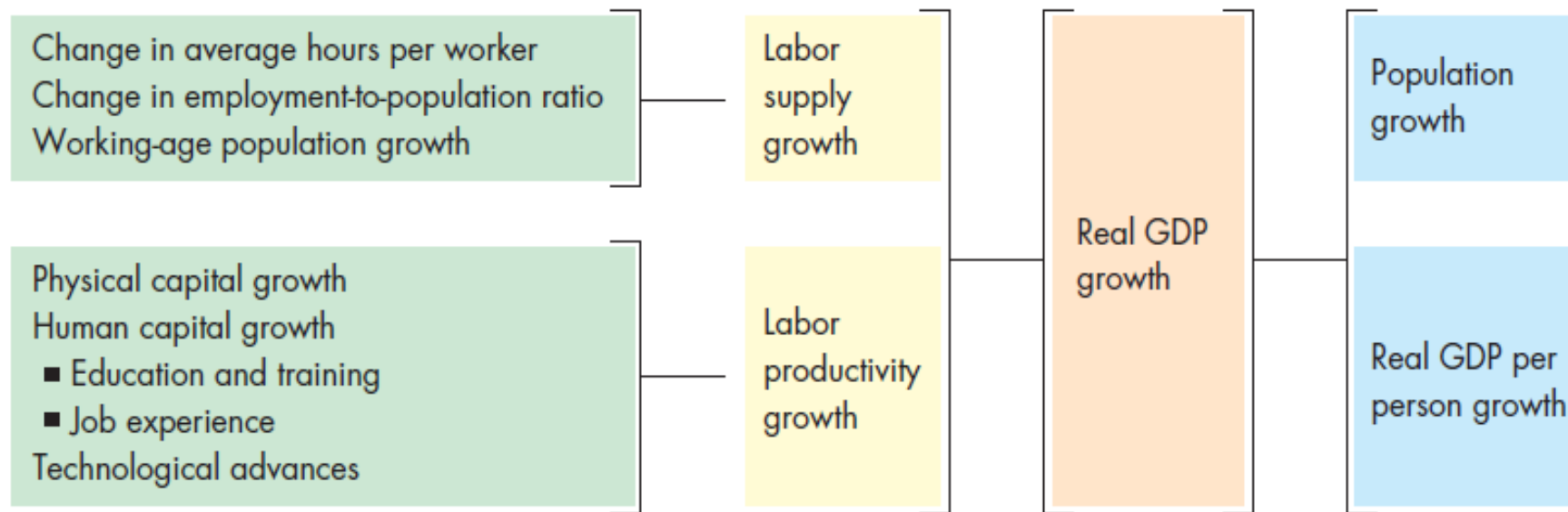
Hence, the increase of capital per worker increases labour productivity

- ***Human capital Growth:*** It is the accumulated skill and knowledge of human beings, is the fundamental source of labour productivity growth

One of the most important form of Human Capital Growth is through the repetition in doing production tasks

- ***Technological Advances:*** Despite the other factors having key contributions towards labour productivity growth, the discovery and the application of new technologies has made an even greater contribution

Preconditions for Labour Productivity Growth



Labor supply growth and labor productivity growth combine to determine real GDP growth. Real GDP per person growth depends on real GDP growth and population growth.

Policies for Achieving Faster Growth

- ***Stimulate saving:*** Saving finances investment so stimulating saving increases economic growth; A key reason for the East Asian Miracle was a high savings rate
Tax incentives are really effective in stimulating savings
According to Economists, a tax on consumption rather than income provides the best saving incentive
- ***Stimulate research and development:*** Everyone can use the fruits of basic research and development efforts.
So the Governments can direct public funds toward financing basic research for achieving faster growth

Policies for Achieving Faster Growth

- ***Improve the quality of education:*** By funding basic education and by ensuring high standards in basic skills such as language, mathematics, and science, governments can contribute to a nation's growth potential.

Education can also be stimulated and improved by using tax incentives to encourage improved private provision

- ***Provide international aid to developing nations:*** Although logic dictates that financial aid from rich nations should boost investment and growth in developing countries, studies show that Aid often gets diverted and spent on consumption, leading to zero and even negative effect
- ***Encourage international trade:*** In practice it is trade, and not aid, that stimulates economic growth. The fastest-growing nations are those most open to trade. If the rich nations truly want to aid economic development, they will lower their trade barriers against developing nations

Policies for Achieving Faster Growth

Influence	Good for Economic Growth	Bad for Economic Growth
Region	Far from equator	Sub-Sahara Africa
Politics	Rule of law	Revolutions
	Civil liberties	Military Coups
		Wars
Economic System	Capitalist	
Market Distortions		Exchange rate distortions
		Price controls and black markets
Investment	Human Capital	
	Physical Capital	
International Trade	Open to trade	

Important Note!!!

***REMEMBER TO GO THROUGH CHAPTER 23 FROM THE BOOK FOR A
BETTER UNDERSTANDING OF THE FACTORS CONTRIBUTING TOWARDS
ECONOMIC GROWTH!!!***